Neural Network and Stack Up Models for Diamond Price Prediction

MSDS 458 Assignment 1 Research Report

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Abstract

This study evaluated fully connected neural networks for predicting diamond sale prices from physical measurements and categorical quality grades. The analysis used the 53,940 row Diamonds dataset and retained 53,771 observations after removing impossible dimensions, extreme geometric values, and exact duplicate records. Numerical predictors were standardized, categorical predictors were one hot encoded, and the resulting 23 features were divided into training, validation, and test sets. Twelve modeling experiments compared single hidden layer networks, deeper networks, raw and log transformed targets, dropout regularization, and two stack up strategies. The best model under the prespecified validation MAPE criterion was a 256 128 64 network trained on log price with 15 percent dropout. On 10,755 test observations, it produced a mean absolute error of $282.08, root mean squared error of $557.60, MAPE of 6.91 percent, and R squared of 0.9800. A total of 95.57 percent of predictions were within 20 percent of actual price, with a 95 percent Wilson interval of 95.17 to 95.95 percent. The model therefore met the coverage requirement but missed the 5 percent average MAPE target. The results show that transforming the skewed target improved percentage accuracy more than increasing depth alone. The model is suitable for expert assisted pricing, but the remaining systematic error does not support fully automated deployment.

Keywords diamond pricing, neural networks, regression, stack up models, MAPE, t SNE

Introduction

Diamond valuation combines measurable physical characteristics with graded quality attributes. The Gemological Institute of America organizes the most widely recognized quality framework around carat weight, color, clarity, and cut. Two diamonds with similar weight can command different prices because the other quality dimensions change rarity and appearance (Gemological Institute of America, 2014). A seller must integrate these attributes consistently while also accounting for nonlinear interactions. Manual appraisal provides domain judgment, but it requires time and can vary across evaluators. A predictive model can supply a repeatable initial estimate for expert review.

The management problem in this study was to estimate the sale price of a new diamond from carat, depth, table, length, width, height, cut, color, and clarity. Management established two performance requirements. First, average test MAPE had to be no greater than 5 percent. Second, at least 95 percent of individual test predictions had to have an absolute percentage error below 20 percent. These requirements emphasize relative accuracy across the price distribution rather than fit alone. A model can explain most price variance and still fail the business target if a material group of inexpensive or unusually expensive diamonds receives poor estimates.

The analysis addressed four questions. The first was whether networks with multiple hidden layers improved performance over single hidden layer networks. The second was whether training on log transformed price reduced the effect of the strongly skewed response. The third was whether a two stage stack up design, consisting of a price tier classifier and tier specific regressors, outperformed a single global model. The fourth was whether the final hidden layer organized diamonds by price more clearly than the observed input features.

The experiment produced a clear operational conclusion. Neural networks modeled the pricing relationship well in terms of explained variance, but the strongest improvement came from changing the target representation rather than adding layers. The selected log price network met the coverage requirement with a small margin and failed the average MAPE target by 1.91 percentage points. This report therefore recommends controlled use as a pricing aid with expert review and continued model development before automated pricing.

Literature Review

The Diamonds dataset distributed with ggplot2 contains prices and nine descriptive attributes for more than 50,000 round cut diamonds (Wickham, 2016). Its fields align closely with the commercial grading framework. Carat measures weight, while cut, color, and clarity represent ordered quality categories. Length, width, height, depth, and table capture geometry. The combination creates a useful tabular regression problem because the predictors include continuous measurements, ordered categories, substantial collinearity, and nonlinear price effects.

Previous analyses of this dataset show that model choice and preprocessing materially change conclusions. Panwala (2019) compared several regression methods and reported that a decision tree performed best in that implementation, followed by neural networks and polynomial regression. Crépin (n.d.) likewise treated diamond pricing as a supervised learning problem with data exploration and model comparison. These course supplied examples establish a useful baseline but do not answer whether a deeper network, a transformed target, or a mixture of classification and regression can satisfy explicit percentage error thresholds.

Peer reviewed studies reach a similar conclusion about the nonlinear nature of diamond pricing. Kigo, Omondi, and Omolo (2023) compared regression and classification algorithms and identified strong performance from random forest models. Their study emphasized that carat, cut, clarity, table, and depth combine nonlinearly, which limits simpler linear specifications. Xiao (2024) compared random forest, k nearest neighbors, XGBoost, and multilayer perceptron models on the same 53,940 row dataset. XGBoost achieved the strongest results in that comparison, with test R squared near 0.982. These findings suggest that high fit is attainable, but they also show that no single algorithm dominates every objective. The present study narrows the question to fully connected neural networks and evaluates them against business thresholds based on MAPE and tail coverage.

The neural network design follows established approximation and optimization results. Hornik, Stinchcombe, and White (1989) showed that feedforward networks with a sufficiently large hidden layer can approximate broad classes of functions. Additional depth may represent interactions more compactly, but the theorem does not imply that deeper networks will generalize better on a particular dataset. The experiments therefore compare width and depth empirically rather than assuming that a larger model must win.

All networks were optimized with Adam, which uses adaptive estimates of gradient moments to update parameters efficiently (Kingma & Ba, 2015). One experiment added dropout. Dropout randomly removes units during training to limit coadaptation and approximate an ensemble of thinner networks, which can reduce overfitting in large networks (Srivastava et al., 2014). The selected model combined dropout with a log transformed response. This transformation is especially relevant because raw diamond prices are right skewed and MAPE gives equal proportional importance to inexpensive and expensive observations.

The hidden representation was evaluated with t distributed stochastic neighbor embedding. t SNE maps high dimensional observations into two dimensions while seeking to preserve local neighborhoods (van der Maaten & Hinton, 2008). The method is appropriate for qualitative comparison of the encoded inputs and the final hidden layer, but distances between remote clusters should not be interpreted as a formal measure of separation. In this study, t SNE serves as visual evidence of what the network learned, not as a model selection metric.

Methods

Dataset and Variables

The response variable was price in US dollars. Six numerical predictors were used: carat, depth, table, x, y, and z. Three categorical predictors represented cut, color, and clarity. The raw dataset contained 53,940 observations, ten columns, and no missing values. Price ranged from $326 to $18,823, with a median of $2,401 and a mean of $3,932.80. The large difference between the mean and median, together with the long upper tail, motivated explicit comparison of raw and log transformed targets.

Table 1

Analysis dataset and partition sizes

Component	Observations	Share	Purpose
Raw data	53,940	100.0%	Initial exploration and cleaning

Component	Observations	Share	Purpose
Clean data	53,771	99.7%	Modeling population
Training	34,412	64.0%	Parameter estimation
Validation	8,604	16.0%	Early stopping and model selection
Test	10,755	20.0%	Final performance assessment

Data Cleaning and Partitioning

Cleaning removed rows with nonpositive x, y, or z dimensions, y or z values above 20 millimeters, and table values of 80 or greater. Exact duplicate rows were also removed. The combined rules removed 169 observations, or 0.31 percent of the original data. Because the dataset does not include unique diamond identifiers, removing exact duplicates assumes that repeated rows represent duplicate records rather than distinct diamonds with identical recorded attributes. This assumption is revisited as a limitation.

The clean data were divided with random seed 42. Twenty percent was reserved for the final test set. The remaining observations were split into training and validation samples, producing an overall 64 16 20 allocation. The test set remained separate during network fitting and early stopping. Model selection used validation MAPE, which matched the primary business objective and avoided selecting a model solely because it performed favorably on the test sample.

Preprocessing

The preprocessing transformer was fitted on the training data. Numerical variables were standardized to zero mean and unit variance. Cut, color, and clarity were one hot encoded with one reference category removed from each feature. The procedure generated 23 model inputs. The fitted transformer was then applied without refitting to the validation and test observations. This sequence prevented information from the validation or test distributions from influencing the learned scaling parameters.

Each response representation was standardized using the training response. Raw target experiments standardized price directly. Log target experiments applied log one plus price before standardization and converted predictions back to dollars with the inverse transformation. The log transformation compressed the long upper tail and changed squared error optimization from emphasizing large dollar deviations toward a more proportional treatment of price differences.

Neural Network Experiments

Ten global neural network experiments and two stack up experiments were completed. The single hidden layer group used 16, 32, 64, or 128 ReLU units. The raw price deep network group used 64 32, 128 64, and 128 64 32 architectures. Three log target experiments evaluated a 128 64 32 model, a lower learning rate version of that model, and a wider 256 128 64 model with 15 percent dropout and small L2 weight decay. All models used a linear output node, mean squared error loss on the standardized target, mini batches, Adam optimization, and early stopping with restoration of the best validation weights.

The architecture tests were intentionally structured rather than treated as an unrestricted search. The single layer runs measured the effect of width. The raw deep runs measured the effect of adding hidden

layers. The log target comparison isolated the effect of transforming the response. The lower learning rate run tested optimization sensitivity, and the dropout run examined whether regularization helped a wider model generalize. Appendix B lists the full configuration and outcomes for all twelve experiments.

Stack Up Modeling

Price tiers were defined from the training response using the 33rd and 67th percentiles. The resulting boundaries were approximately $1,235 and $4,285, producing 11,486 low, 11,455 medium, and 11,471 high training observations. A 128 64 32 classifier with 10 percent dropout predicted the tier from the same 23 encoded inputs. Three separate 128 64 32 log price regressors were trained using observations from their assigned tiers.

Two combination rules were evaluated. Hard routing selected the regression output associated with the classifier's most probable tier. Soft weighting generated a prediction from each tier regressor and averaged those predictions using the classifier probabilities. Soft weighting was expected to reduce discontinuities around tier boundaries because an observation near a cutoff could receive information from both adjacent regression models.

Model Evaluation

Performance was summarized with mean absolute error, root mean squared error, MAPE, R squared, the 95th percentile of individual absolute percentage error, and the percentage of predictions below 20 percent error. MAPE was computed as MAPE = (100 divided by n) times the sum of the absolute value of actual price minus predicted price divided by actual price. The coverage requirement used the indicator that individual absolute percentage error was below 20 percent. A 95 percent Wilson interval was calculated for the selected model's observed coverage proportion.

The primary selection criterion was validation MAPE because management specified an average percentage error target. Test performance was reported only after selection. The analysis also retained MAE, RMSE, and R squared because a model selected on relative error may not minimize dollar error or large residuals. This distinction proved important when comparing the selected global DNN with the soft stack up model.

Results

Exploratory Findings

The raw price distribution was strongly right skewed. Most diamonds were priced below $5,000, while a thinner tail extended beyond $18,000. The log transformation spread the lower price observations and compressed the upper tail, producing a response that was more suitable for proportional prediction. Carat had a correlation of 0.92 with price. The three dimensions were also strongly correlated with carat and with one another, reflecting their common relationship to stone size. Depth had almost no linear association with price, while table had a weak positive association.

The categorical boxplots require care. Premium and fair diamonds showed higher unadjusted median prices than ideal diamonds, and lower color or clarity grades sometimes had higher raw median prices. These patterns do not imply that lower quality increases value. Larger stones are unevenly distributed across the categories, so carat confounds the marginal comparisons. The neural network was intended to learn the joint relationship rather than interpret any one boxplot causally.

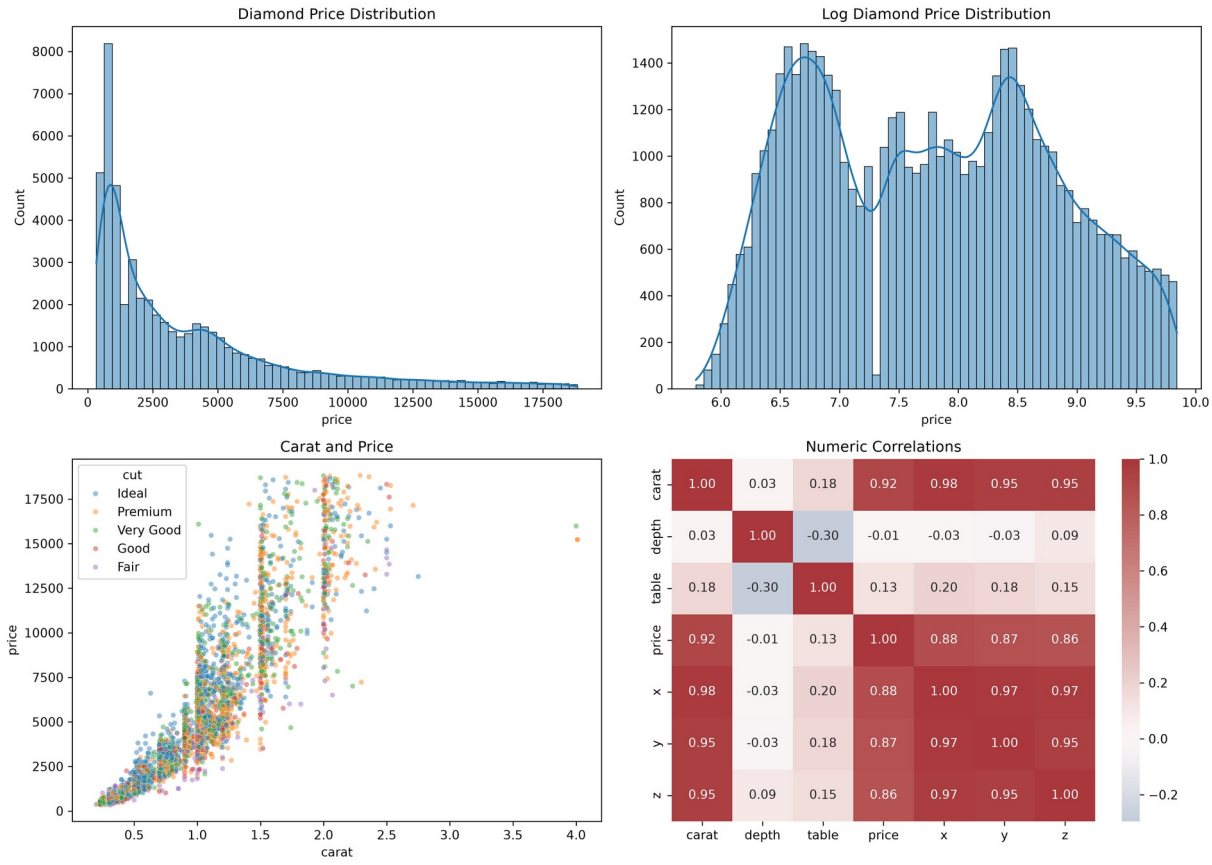


Figure 1 *Price distribution carat relationship and numerical correlations*

Source Author analysis of the Seaborn Diamonds data

Model Comparison

The raw target networks produced test MAPE values from 9.13 to 9.86 percent. Increasing width within the single hidden layer group did not yield a consistent improvement, and adding depth alone reduced MAPE only modestly. The 128 64 32 raw target DNN was the strongest raw model on MAPE at 9.13 percent, with 90.79 percent of its test predictions below 20 percent error.

Changing the target to log price produced the largest improvement. The 128 64 32 log model reduced test MAPE to 6.98 percent and increased coverage to 95.83 percent. Relative to the same raw architecture, this was a 24.4 percent reduction in MAPE. Lowering the learning rate to 0.0003 worsened test MAPE to 7.22 percent and required 79 epochs, suggesting slower or incomplete optimization under the 80 epoch limit.

The selected model was E10, the 256 128 64 log price DNN with 15 percent dropout. It had the lowest validation MAPE at 6.77 percent and achieved test MAPE of 6.91 percent. Its test MAE was $282.08, RMSE was $557.60, and R squared was 0.9800. The 128 64 32 log model produced slightly better MAE, RMSE, R squared, and coverage on the test set, but it was not selected because its validation MAPE was marginally higher. This result illustrates why the decision rule must be specified before reviewing test outcomes.

Table 2

Performance of the principal comparison models

Model	Test MAPE	Within 20%	MAE	RMSE	R squared
E07 raw DNN	9.13%	90.79%	$301.62	$553.96	0.9803
E08 log DNN	6.98%	95.83%	$279.56	$531.80	0.9818
E10 log DNN dropout	6.91%	95.57%	$282.08	$557.60	0.9800
E11 hard stack up	7.11%	95.01%	$279.32	$537.48	0.9815
E12 soft stack up	6.94%	95.47%	$274.75	$534.00	0.9817

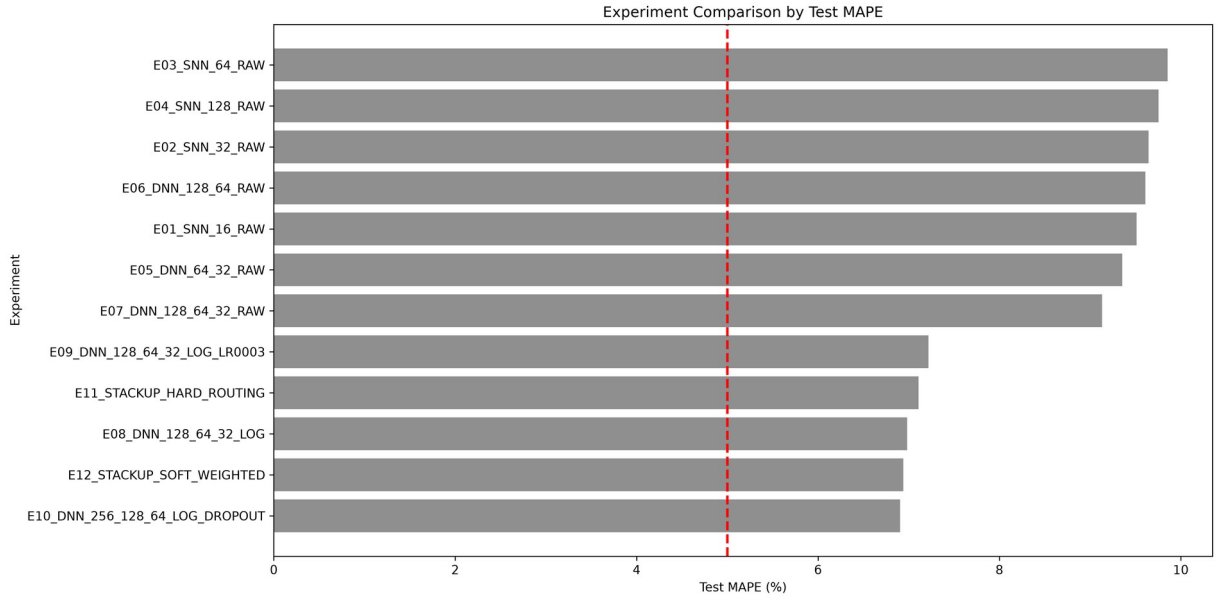


Figure 2 *Test MAPE for all neural network and stack up experiments*

Source Author analysis of the Seaborn Diamonds data

Stack Up Results

The price tier classifier achieved validation accuracy of 95.73 percent and test accuracy of 95.43 percent. Its errors were local to neighboring tiers. No low price test observations were classified as high, and no high price observations were classified as low. This pattern shows that the classifier captured broad price regions and that its mistakes concentrated near the learned boundaries.

Soft weighting improved on hard routing. Hard routing produced test MAPE of 7.11 percent and MAE of $279.32. Soft weighting reduced MAPE to 6.94 percent and MAE to $274.75. The soft model had the lowest dollar MAE of the twelve experiments, but its validation MAPE of 6.79 percent remained slightly above E10's 6.77 percent. The global dropout DNN was therefore selected for the stated percentage error objective. If management instead prioritized average dollar error, the soft stack up would be a reasonable alternative.

Business Requirements

The selected model failed the average MAPE requirement. Its 6.91 percent test MAPE exceeded the 5 percent target by 1.91 percentage points. It passed the individual error requirement: 10,279 of 10,755 test predictions, or 95.57 percent, had absolute percentage error below 20 percent. The 95 percent Wilson interval for this proportion was 95.17 to 95.95 percent, so its lower confidence bound also exceeded 95 percent. The 95th percentile of individual absolute percentage error was 18.88 percent.

Table 3

Selected model evaluation against management requirements

Requirement	Target	Observed	Decision
Average percentage error	MAPE no greater than 5%	6.91%	Not met
Individual error coverage	At least 95% below 20% error	95.57%	Met
Coverage confidence interval	Lower bound at least 95%	95.17% to 95.95%	Met
Tail percentage error	95th percentile below 20%	18.88%	Met

Residual and Tier Behavior

Prediction errors were heteroscedastic in dollar terms. Residual spread increased with predicted price, and the actual versus predicted plot showed compression among the most expensive diamonds. Several high price observations were underpredicted, including one large miss near the upper end of the test range. This pattern explains why RMSE remained nearly twice MAE even though overall R squared was high.

Percentage performance was weakest in the medium tier. E10 achieved MAPE of 6.56 percent for low price diamonds, 7.18 percent for the medium tier, and 6.97 percent for the high tier. Coverage below 20 percent error was 97.58 percent, 93.86 percent, and 95.32 percent, respectively. Further research should therefore focus on observations around $1,235 to $4,285 and on outliers among expensive stones. Complete residual and tier results appear in Appendices C and D.

Hidden Layer Representation

The input t SNE map contained mixed price colors within many local groups. The final hidden layer produced a much clearer progression from low to high log prices across several learned branches. The network therefore transformed the 23 encoded predictors into features that were more directly ordered by price. This visual pattern supports the claim that the hidden nodes learned pricing related structure rather than merely reproducing the input geometry.

The t SNE result remains qualitative. The algorithm preserves local neighborhoods more reliably than global distances, and different random seeds can change the visible orientation. It does not prove that two distant branches represent distinct economic regimes. Its value here is comparative: the same sampled diamonds display a clearer price gradient after the network's transformation than before it.

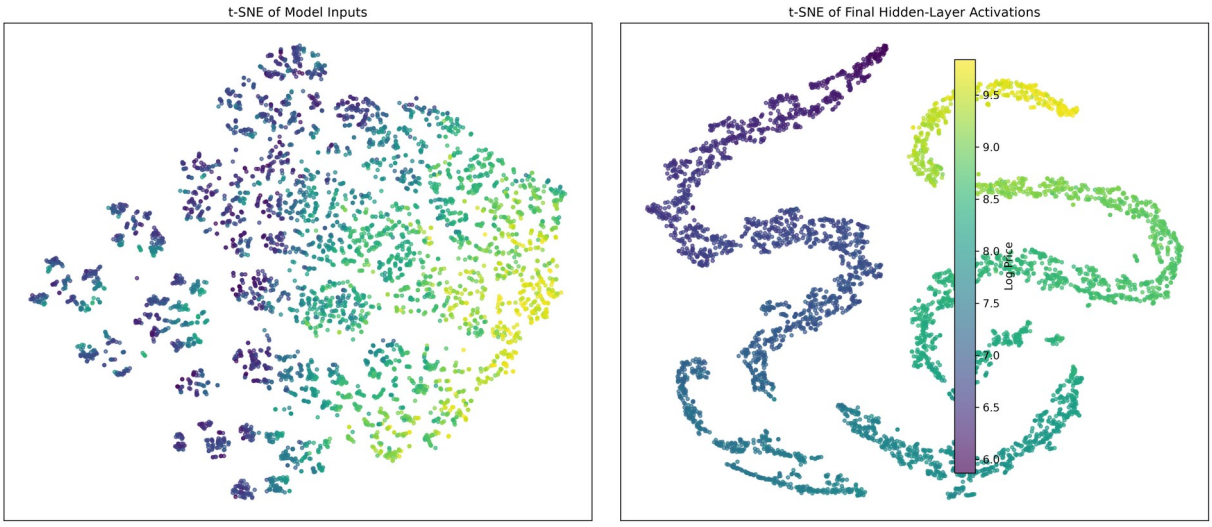


Figure 3 *t SNE comparison of encoded inputs and final hidden layer activations*

Source Author analysis of the Seaborn Diamonds data

Limitations

The dataset has several limitations for operational deployment. It describes historical round cut diamonds but does not include seller, transaction date, location, certification laboratory, fluorescence, shape beyond the dataset's scope, or changing market conditions. A model trained on these records estimates the conditional pattern in the dataset, not a current universal market price. Its predictions should be recalibrated before use with new inventory or a different sales channel.

The cleaning decision to remove exact duplicate rows is also uncertain because the source lacks unique identifiers. Two separate diamonds can share the same rounded attributes and price. Removing duplicates prevents identical records from receiving extra weight, but it may also remove legitimate observations. A sensitivity analysis that retains repeated rows would clarify the impact.

Only fully connected neural networks and the specified stack up design were evaluated. Prior research often finds strong results from random forest and gradient boosting models (Kigo et al., 2023; Xiao, 2024). Those models were outside the assignment's backpropagation focus, but they remain relevant benchmarks for a production decision. Finally, the experiment used one fixed split. Repeated cross validation or repeated train validation splits would provide a stronger estimate of variation across samples.

Conclusions

The experiment answered the four research questions. Deeper raw price networks slightly improved percentage error over most single hidden layer models, but depth did not produce a decisive change. Log transforming price produced the major improvement, lowering MAPE from 9.13 percent to approximately 7 percent for the comparable 128 64 32 network. The soft stack up reduced dollar MAE relative to the selected model but did not improve validation MAPE. The hidden layer t SNE map showed a clearer price gradient than the encoded inputs, indicating that the network learned a pricing oriented representation.

The selected E10 network explained approximately 98 percent of test price variance and produced an average absolute dollar error of $282.08. Those statistics indicate strong overall fit, but they do not override the business criterion. Its 6.91 percent MAPE did not meet the 5 percent requirement. In contrast, 95.57

percent of test predictions were within 20 percent error, and the confidence interval supported that coverage threshold. The best model is therefore useful but not sufficient for fully automated pricing.

Management Recommendation

Management should deploy the E10 model as an appraisal support tool. The system can produce an initial estimate and flag cases for expert review. Expert review should remain mandatory for diamonds in the medium price tier, observations with unusual combinations of size and quality, and high value stones where dollar residuals widen. The model should not issue an unreviewed final sale price because the primary 5 percent MAPE objective was not achieved.

The next modeling cycle should test a percentage aligned loss function, richer interaction features, repeated validation splits, and an ensemble that combines the global log model with the soft stack up. The business should also collect transaction date, certification, fluorescence, shape, seller, and market context. Progress should be judged against the same untouched test protocol, with separate monitoring of low, medium, and high price segments. Automated pricing should be reconsidered only after average MAPE reaches 5 percent without sacrificing the demonstrated 95 percent coverage.

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Appendix A Exploratory Data Analysis

Appendix A contains supporting distributions used to assess categorical predictors and cleaning decisions. The plots display unadjusted prices and should be interpreted jointly with carat and the other predictors.

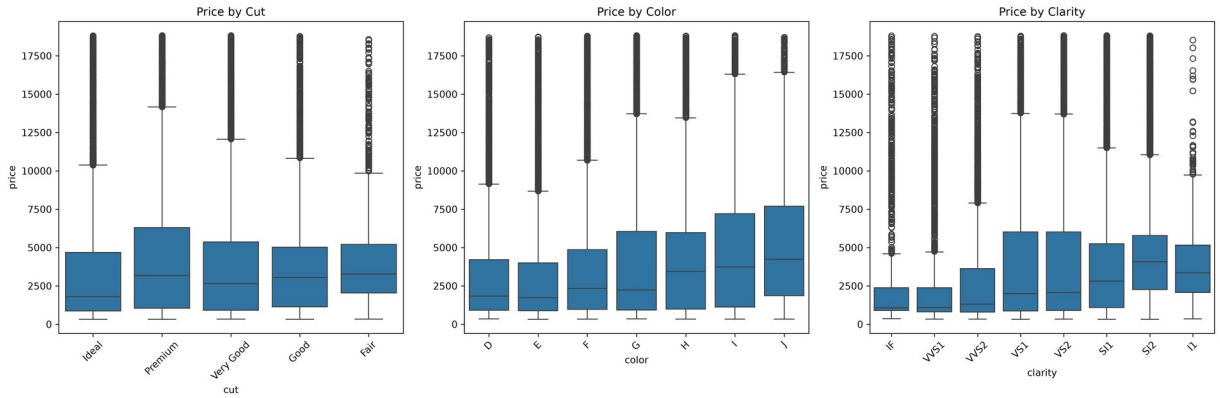


Figure A1 *Unadjusted price distributions by cut color and clarity*

Source Author analysis of the Seaborn Diamonds data

Table A1

Data cleaning rules

Cleaning rule	Rationale	Effect
x y and z greater than zero	Removes physically impossible dimensions	Part of 169 total removals
y and z below 20	Removes extreme geometric values	Part of 169 total removals
table below 80	Removes extreme table value	Part of 169 total removals
Exact duplicate row	Avoids overweighting identical records	146 duplicate rows identified

Appendix B Model Experiments

Table B1

Neural network and stack up configurations

Experiment	Hidden layers	Target	LR	Batch	Dropout	Best epoch
E01 SNN 16 RAW	16	raw	0.0010	256	0.00	73
E02 SNN 32 RAW	32	raw	0.0010	256	0.00	76
E03 SNN 64 RAW	64	raw	0.0010	256	0.00	50
E04 SNN 128 RAW	128	raw	0.0010	256	0.00	62
E05 DNN 64 32 RAW	64 32	raw	0.0010	256	0.00	37
E06 DNN 128 64 RAW	128 64	raw	0.0010	256	0.00	39
E07 DNN 128 64 32 RAW	128 64 32	raw	0.0010	256	0.00	17
E08 DNN 128 64 32 LOG	128 64 32	log	0.0010	256	0.00	61
E09 DNN 128 64 32 LOG LR0003	128 64 32	log	0.0003	256	0.00	79
E10 DNN 256 128 64 LOG DROPOUT	256 128 64	log	0.0010	256	0.15	59
E11 STACKUP HARD ROUTING	Classifier and three regressors	log	0.0010	256	0.10	39
E12 STACKUP SOFT WEIGHTED	Classifier and three regressors	log	0.0010	256	0.10	39

Table B2

Complete test performance by experiment

Experiment	MAE	RMSE	MAPE	R squared	P95 APE	Within 20%
E10 DNN 256 128 64 LOG DROPOUT	282.08	557.60	6.91%	0.9800	18.88%	95.57%
E12 STACKUP SOFT WEIGHTED	274.75	534.00	6.94%	0.9817	19.19%	95.47%
E08 DNN 128 64 32 LOG	279.56	531.80	6.98%	0.9818	18.71%	95.83%
E11 STACKUP HARD ROUTING	279.32	537.48	7.11%	0.9815	19.99%	95.01%
E09 DNN 128 64 32 LOG LR0003	282.16	536.01	7.22%	0.9816	19.35%	95.47%

Experiment	MAE	RMSE	MAPE	R squared	P95 APE	Within 20%
E07 DNN 128 64 32 RAW	301.62	553.96	9.13%	0.9803	24.42%	90.79%
E05 DNN 64 32 RAW	293.52	535.20	9.35%	0.9816	25.99%	89.69%
E01 SNN 16 RAW	308.70	565.69	9.51%	0.9795	25.89%	89.27%
E06 DNN 128 64 RAW	293.84	537.28	9.61%	0.9815	27.60%	87.36%
E02 SNN 32 RAW	303.39	546.32	9.64%	0.9808	25.84%	89.26%
E04 SNN 128 RAW	304.84	544.98	9.75%	0.9809	28.19%	88.07%
E03 SNN 64 RAW	304.01	548.93	9.86%	0.9807	28.92%	87.45%

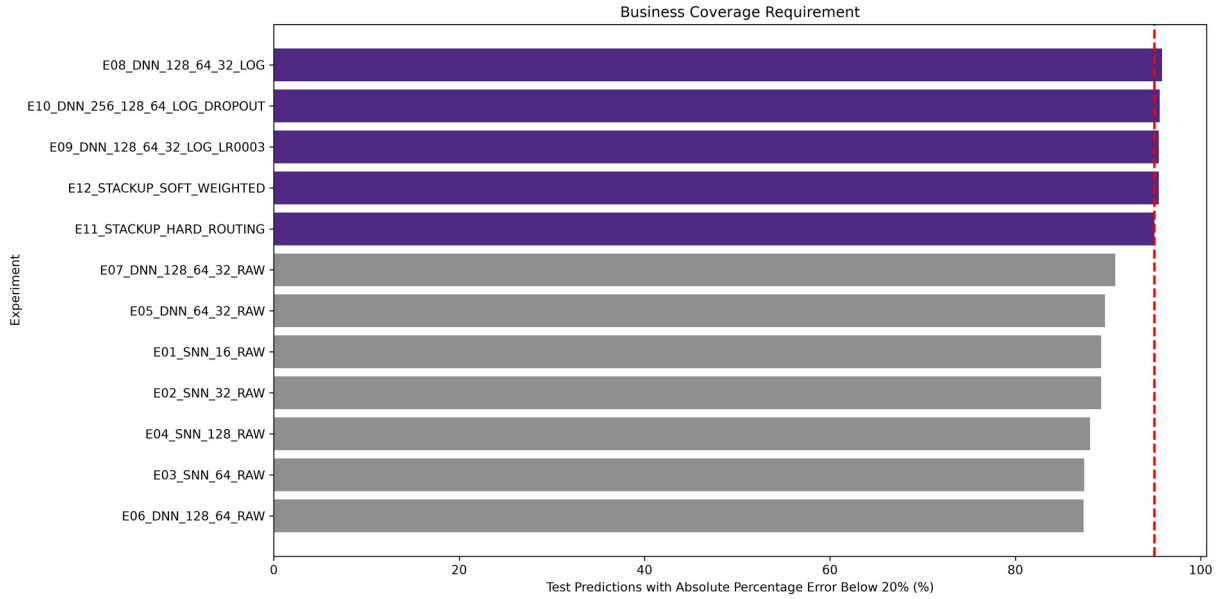


Figure B1 *Percentage of test predictions below 20 percent absolute error*

Source Author analysis of the Seaborn Diamonds data

Appendix C Best Model Diagnostics

The selected E10 model stopped at epoch 59. The validation curve remained stable while the training curve was higher because dropout was active during training and disabled during evaluation. Residual dispersion increased with price, and a small number of observations produced large dollar errors.

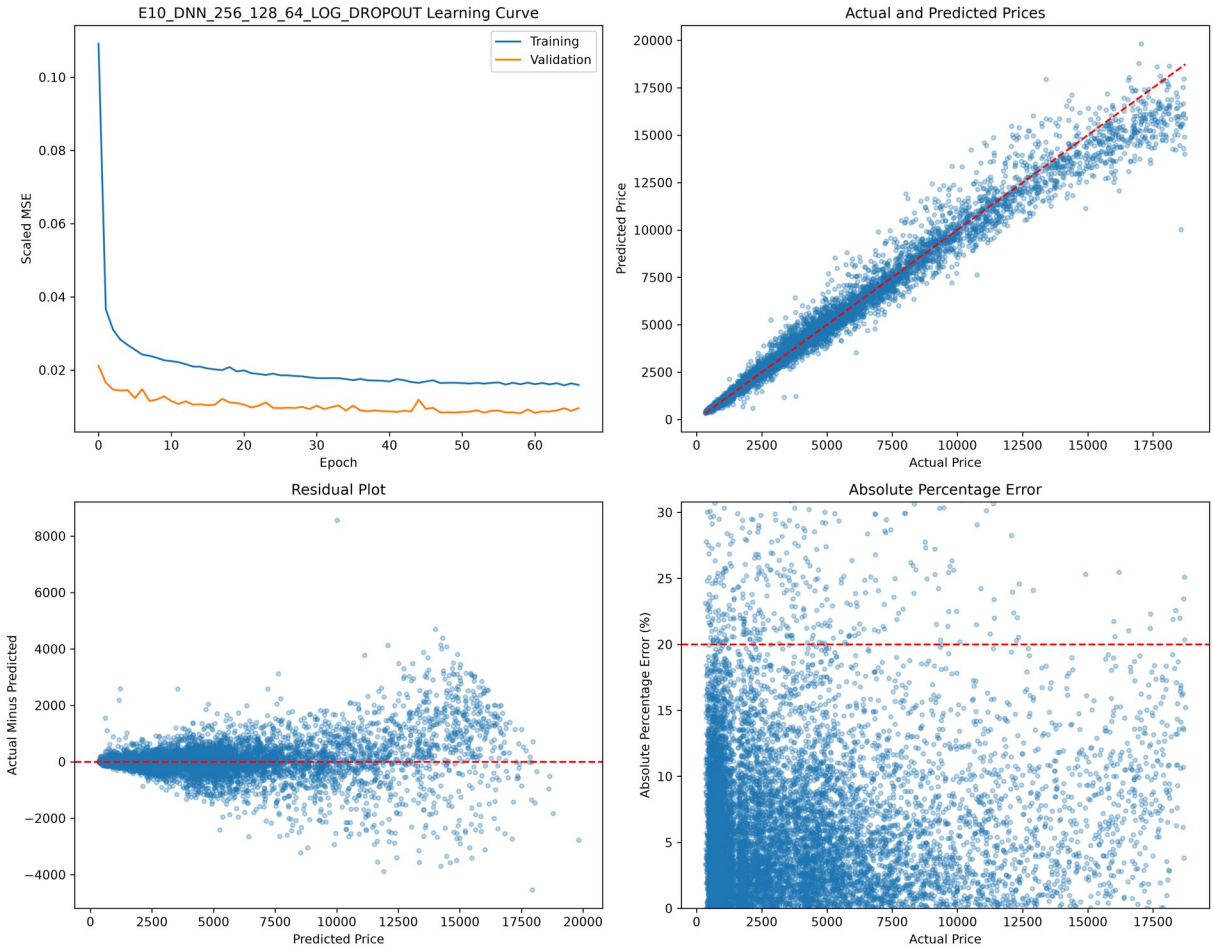


Figure C1 *Learning curve prediction fit residuals and percentage errors for E10*

Source Author analysis of the Seaborn Diamonds data

Appendix D Stack Up Model

The classifier separated the balanced low medium and high tiers with 95.43 percent test accuracy. All classification mistakes occurred between adjacent tiers.

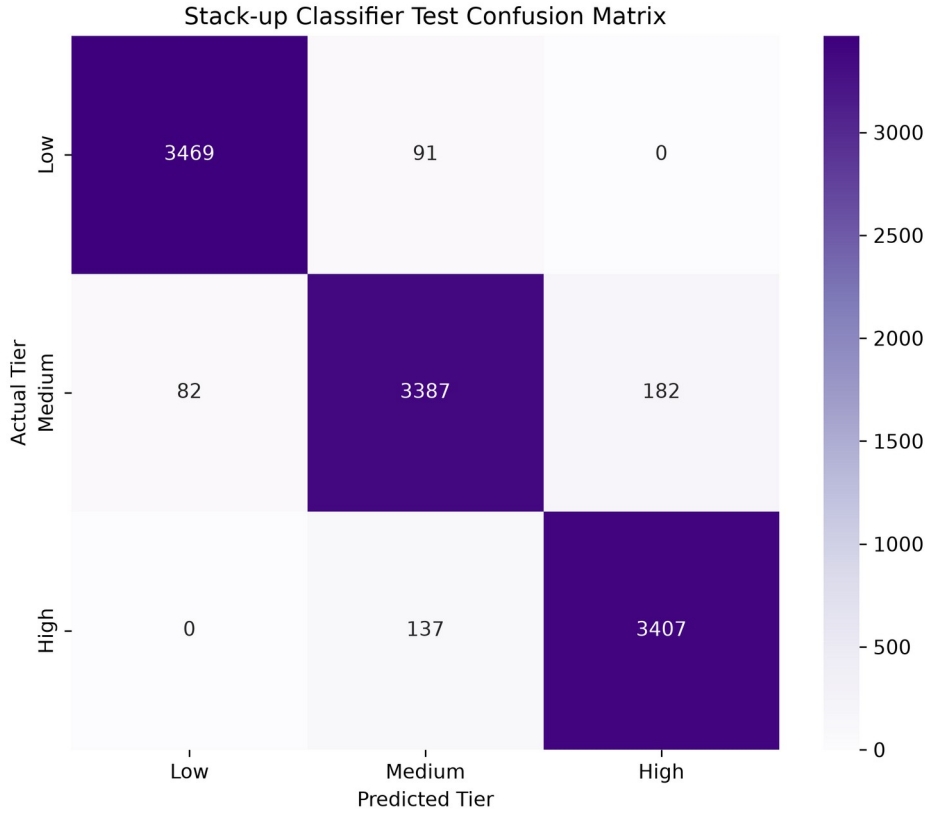


Figure D1 *Test confusion matrix for the price tier classifier*

Source Author analysis of the Seaborn Diamonds data

Table D1

Tier performance for the selected global model and stack up models

Model	Price tier	MAE	MAPE	Within 20%
E10 log DNN	Low	$50.67	6.56%	97.58%
E10 log DNN	Medium	$187.52	7.18%	93.86%
E10 log DNN	High	$611.94	6.97%	95.32%
E11 hard stack up	Low	$53.61	6.87%	96.35%
E11 hard stack up	Medium	$197.39	7.48%	93.89%
E11 hard stack up	High	$590.47	6.97%	94.81%
E12 soft stack up	Low	$51.57	6.71%	96.80%
E12 soft stack up	Medium	$189.68	7.25%	94.55%
E12 soft stack up	High	$586.56	6.86%	95.09%

Appendix E Reproducibility

The analysis used random seed 42 and executed on Apple Metal Performance Shaders. All 21 notebook cells completed without errors. The notebook produced the model weights, validation and test predictions, experiment tables, and figures used in this report.

Table E1

Software and execution settings

Item	Value
Python	3.14.4
PyTorch	2.14.1
NumPy	2.5.3
pandas	3.0.6
scikit learn	1.9.1
seaborn	0.13.2
Random seed	42
Execution device	Apple MPS
Encoded features	23
Maximum epochs	80 for global models and 100 for tier regressors
Early stopping	Validation loss with restored best weights

The executed notebook is submitted separately in HTML format. The main output files include experiment_results.csv, results_by_price_tier.csv, test_predictions.csv, validation_predictions.csv, best_base_model.pt, and stackup_models.pt.